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Optimal return policy and modular design for build-to-order products

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Abstract

A build-to-order product gives the firm the opportunity to customize the product to the requirement of the customers. The firm will see an increase in demand, but face some operational difficulties. One of these is related to the return policy which is now a recognized tool to win customer orders. This is even more true for increasingly popular Internet sales where the opportunity to physically examine the product is absent. A tremendous advantage can be gained by the Internet firm if it could offer a return policy for BTO products. We propose the use of modularity in the product design as a solution to this problem. From manufacturer's point of view, following a policy of modularization and offering a generous return policy would increase revenue, but also increase the cost due to increased likelihood of return and increased cost of design. We develop a profit maximization model to jointly obtain optimal polices for return policy and modularity level in terms of certain market reaction parameters. We obtain a number of managerial guidelines for using marketing and operational strategy variables to influence those reaction parameters so as to obtain the maximum benefit from the market.
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1. Introduction

The last decade has seen phenomenal changes in the market. Two of those changes have engendered a paradigm shift in the market place. One is that customers are demanding varied characteristics in the product offering like customized products and generous return policy. The second phenomenal change is the B-to-C e-Business model for selling directly to the customers bypassing the traditional distributor – wholesaler – retailer chain. These two changes together put the manufacturer in a difficult situation. A customized product needs to be built-to-order (BTO) increasing both lead time and cost of production. A BTO product is essentially built to a particular customer's requirement and, therefore, returning the item is generally not allowed. It is thus possible to gain tremendous competitive advantage if a return policy could be offered for a BTO product, even more so for the direct channel because the

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customers are not able to inspect the item before ordering. In this paper, we will introduce a business model that will allow a firm to offer the customer a BTO product at reasonable cost, and simultaneously offer a return policy without having to absorb the full cost of the BTO product in case it is returned.

As mentioned earlier, return policy has emerged as an important competitive weapon in the marketplace substantially influencing product sales. Many recent studies (see Pinkerton, 1997; Trager, 2000; Wood, 2001) have shown that customers view the ease of return as one of the main motivators for the buying decision and are very likely to consider such policy. The type of return policy practiced in business varies across industry and stores. This policy may be defined in terms of the percentage of selling price returned and also by the level of difficulty in returning the item. The return policy can be in the form of unconditional money back guarantee, or store credit only, or no refund whatsoever. Restrictions imposed by a retailer for returning include short time limits for returning the product, unused product, returned in original packaging, and special instruction on labeling. For example, Dell Computer allows return on all new hardware (including customized order), accessories, parts, and unopened software still in its sealed package within 30 days from the date on the invoice while Apple Computer only allows return within 14 days for most of the products and no return for special and custom orders. These examples show that even in the same industry, return policy can widely vary among the competitors. It also shows that, at least in one case, the firm allows return of BTO products. Researchers agree that e-Business sales would increase at an even faster rate in the presence of a viable and clear return policy (Rogers and Tibben-Lemke, 1999). Surveys of existing return policy can be found in Hess and Mayhew (1997), Davis et al. (1998) and Rogers and Tibben-Lemke (1999).

A generous return policy, however, would increase the probability of return and thereby increase the cost of doing business. The Spiegel Group shipped $1.5 billion of merchandise in 2001 from its Spiegel, Eddie Bauer, and Newport News catalogs and Web sites and saw $300 million in returns (Trebilcock, 2002). Padmanabhan and Png (1995, 1997) demonstrate the strategic role of an unlimited full return policy showing that manufacturers should accept returns only under certain demand and cost conditions. The cost of returned merchandise can be as high as 50% of total sales (Rogers and Tibben-Lemke, 1999). A generous return policy, therefore, would affect both sides in the profit equation. In case of a BTO product, the firms are reluctant to offer a return policy because the returned merchandise is practically useless as it was designed to meet the requirement of a particular customer. Davis et al. (1998), for example, show that full return policy should not be resorted to if high salvage value cannot be realized for the returned merchandise. In other words, return policy will not be suitable in case of a BTO product with almost zero salvage value.

2. Build-to-order, modularity, and return policy

In the previous section, we discussed the difficulty of additional lead time and cost increase faced by firms offering a BTO product and also the difficulty of the firm's offering a return policy in such a scenario. In this section, we introduce the concept of modularity as a solution for both these problems. Modularity is a concept in product design where the final product is built from a number of product modules. This needs innovative product design concepts where the different, almost self-contained, modules are assembled to configure a large number of variations the final product (Sanchez, 2002; Baldwin and Clark, 1997). Chakravarty and Balakrishnan (2001) found that modular design of products is one way to achieve higher product performance without increasing manufacturing cost in a disproportionate manner.

Modularity is defined as a continuum describing the degree to which a system's components can be separated and recombined, and it refers both to the tightness of coupling between components and the degree to which the rules of the system architecture enable the mixing and matching of components (Schilling, 2000). For discussion on the potential benefits and costs from applications of modularity, refer to Mikkola and Gassmann (2003). Benefits of modularity include speeding up of the product development process, increasing the range of product variations, rapid technological upgrading of products, reducing the number of suppliers, and reducing costs of development and production (Sanchez, 2002; Mikkola and Gassmann, 2003). Examples can be

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seen in computers, bicycles, wiring business, among others.

Relevant to our paper, modularity also offers another unique benefit to the firm. We have shown in the previous section that return policy is an important tool to increase sales, more so for online shopping. But a BTO firm is at a disadvantage because of the almost uselessness of the returned product which was built to only a particular customer's specification. In this paper, we propose that modularity in the product design can be used as a solution to the problem. For example, when a BTO product with modular design is returned, the product can then very easily be dismantled (just as it was easily assembled) producing a number of components which are standard products keeping their full value. A returned product then would give back a large salvage value to the firm, thereby cutting down its loss due to the return. It is also clear that the higher the degree of modularity in the product, the higher is the reduction in the loss. In other words, the company will incur a lower cost of return from the return product when the level of modularity is higher. Some authors (for example, Murray and Sako, 1999) point out that introducing modularity might do one of two things: make the product less distinct or make the value migrate to the module supplier. This usually occurs when the modules are outsourced. The classic example from the computer industry shows that Microsoft sells the same operating system, Windows XP, to IBM and HP, making the computer less distinct. As Murray and Sako (1999) pointed out, there are two distinct stages in the shift towards modular products, for design and production, and for outsourcing. In our paper, we assume that everything remaining constant, the design modularity will increase the demand for the product for the reasons given earlier. Whether or not it will improve the profit will depend on many other factors. Our model optimizes the profit function. The second distinct advantage of modularity for the firm is reduction in costs (e.g. Kim and Chhajed, 2000; Ulrich, 1995).

The specific questions we will address in this paper include the definition of a generous return policy, optimal return policy, level of modularity and its role in the return policies of BTO products, and optimum level of modularity, for a profit maximizing firm. We will present a definition of the degree of modularity and a model to yield the optimal policies regarding return and the design modularity. The model will include a suitable demand function and the relevant costs.

The rest of this paper is organized as follows. In the following section, we will introduce our profit maximizing model and its optimality conditions. We will also show the interactions among the variables such as return policy, level of modularity and profit in the presence of various market reaction parameters. In Section 4, we present our main results on optimal policies and sensitivity analyses. In Section 5, we present the results of our numerical analysis to provide further insights into optimal policies. Section 6 concludes the paper with a summary of managerial implications and suggestions for future research.

3. Model formulation

We consider a supply chain system consisting of two parties, a manufacturer who builds and sells a product online and customers who buy the product. Our model, however, would be applicable to a traditional manufacturer of brick and mortar kind as well. We formulate the flow of payment as follows. A customer orders a product to a certain specification; the manufacturer builds the product to order from standard components and sells at a price p. In case, the product does not meet the customer's expectations, she may decide to return the product. The seller will give r dollars back to the customer as the refund amount where $0 \leq r \leq p$. A higher r means a more generous return policy. In our model, a second decision variable for the firm is the level of modularity that the firm is designing into the product. We denote the degree of modularity in the product design by m. A higher m means a higher level of modularity.

There are a number of measures of modularity in the literature. Mikkola and Gassmann (2003) propose a modularization function which is a function of the total number of components in the product, the number of components dedicated to the product, the degree of coupling (which depends on the number of interfaces between the components), and the substitutability factor (the number of product families made possible by the number of interfaces). Ger-

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shenson et al. (1999) gives a four-step process to measure the modularity in a product. It takes into account the dependency and similarity of components and processes. According to Balachandra (2002), modularity of a product can be measured by the effort involved in the redesign of the product. The product's modularity is the weighted sum of modularity of all the components in the product. Thus the product's modularity is a function of how the components are related to each other so that the effort (and cost) of designing/redesigning the product governs the modularity of the product.

In our model, the variable m can be operationalized using any of the measures given above. Basically, our modularity m improves the attractiveness of the product by satisfying the individual customer's requirement. At the same time, the modularity would help the firm recover a part of its cost by disassembling (with less effort or cost) the returned product into a number of fully reusable standardized components.

3.1. The demand function

Consistent with the objective of our paper, we assume that a generous return policy offered by the seller will generate higher demand. Similarly tightening return policy would decrease demand. Reda (1998) cites Best Buy tightening their return policy (decreasing r) only to find that demand decreased. Davis (2001) believes that minor adjustments to a seller's business strategy such as a simple return policy can increase sales. At the same time, we also assume that a higher level of modularity will generate higher demand because the product can be customized enough to meet the customer's requirements. Product modularity increases demand in several ways. It enables customers to extend the life of the product and minimize the maintenance and repair costs (Sako, 2004). It gives customers the flexibility to tailor the product according to their needs hence it increases customer satisfaction and therefore customer retention (see Murray and Sako, 1999). Product modularity also enables firms to offer more product variety, more new model introduction, and more new products introduced in a given time interval (Brown and Bessant, 2003; Tu et al., 2004). Lastly, a higher price is assumed to have a negative impact on the demand. The demand for the product D, thus, is a function of p, r, and m:

$$D = f(p, r, m) \tag{1}$$

with

$$\frac{\partial D}{\partial p} < 0, \qquad \frac{\partial D}{\partial r} > 0 \quad \text{and} \quad \frac{\partial D}{\partial m} > 0 \tag{2}$$

Without any loss of generality, we assume a linear demand function as used by many researchers in this area (e.g. Tsay and Agrawal, 2000; Padmanabhan and Png, 1997). The linear demand equation will help us get the insights from the model vis-à-vis optimal policies instead of intractable equations leading to no solutions. We will thus assume that the demand of the product will take the following form:

$$D = \alpha - \beta p + \gamma r + \delta m \tag{3}$$

The parameters α, β, γ, and δ, all >0, are explained as follows. α represents the primary demand which depends on factors such as product quality, brand image, and general economic factors outside the scope of this paper. β is the sensitivity of the demand with respect to price. Specifically, as p increases, demand is reduced from its base value at the rate of β. γ is the sensitivity of demand with respect to the return policy and represents the rate of demand increase from the base value as return policy become more generous (r increases). Similarly, δ is the gain in demand with respect to the level of modularity.

3.2. The return quantity function

In our model, the seller allows the customer to return the item for a refund of r dollars. While this policy will motivate more demand (see Eq. (3)), this will also generate more quantity returned by the customer. We model this by the following linear equation:

$$R = \phi + \psi r \tag{4}$$

where R is the returned quantity. Parameter $\psi > 0$ is the rate of return with respect to the refund amount r motivating more people to return because returning the item becomes more and more worthwhile. $\phi > 0$ is a base return quantity which depends on factors other than the refund amount. As seen in Eq. (4), we assume $\partial R/\partial r > 0$.

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3.3. The cost function

We use two types of costs in our model. First, denoted by C_1, is the cost associated with design efforts to increase the modularity level by m. Increasing design modularity is not trivial. Firms need to invest in acquiring the relevant designing skills and appropriate technology to be able to increase modularity. This cost is thus assumed to be increasing and convex in m. We use the following functional form:

$$C_1 = \frac{1}{2}\theta m^2 \tag{5}$$

The second cost, denoted by C_2, is the value lost per unit of the product that has been returned by the customer. To characterize C_2, we assume that the maximum amount of loss for a returned item is r, the amount actually returned to the customer. This assumes that nothing can be salvaged from the returned product. Note that, when the product is modularized, it can be disassembled into a number of standard components which can then be reused giving some salvage value to the firm in effect reducing C_2. The higher the m, the higher is this reduction. We, therefore, assume that C_2 will have the following function form:

$$C_2 = r - \upsilon m \tag{6}$$

Note that higher level of modularity means a larger value that can be recovered from the returned merchandise, reducing C_2 from the maximum value of r.

3.4. Modeling the profit function

The seller's profit function is

$$\pi = pD(p,r,m) - rR(r) - C_1(m) - C_2(r,m)R(r) \tag{7}$$

where pD is the total revenue obtained by selling D units at a price p per unit, rR is the total amount refunded to customers, C_1 is the fixed cost of increasing the level of modularity, and C_2R is the total loss of value from the returned merchandises that cannot be recovered. Note that the total revenue is net of production cost. We treat price as an exogenous variable.

Plugging in the expression for D, R, C_1, and C_2 from Eqs. (3)–(6) into (7), we get the expression for profit as a function of p, r, and m and the sensitivity parameters:

$$\pi = (\alpha - \beta p + \gamma r + \delta m)p - (\phi - \psi r)(2r - \upsilon m) - \frac{1}{2}\theta m^2 \tag{8}$$

As can be seem from Eq. (8), the effect of an increasing r and/or m is not obvious because it increases one term in the profit function but reduces the other. The objective of the firm would be, therefore, to decide on the optimal return policies and optimal level of modularity that maximize profit.

4. Analytical results and sensitivity analyses

In this section, we will use our model and its optimality conditions to derive a number of propositions giving insights to the optimal policies regarding return policy and modularity in design. We also recognize that these results will be predicated on the relative values of the sensitivity parameters in our model. We will, therefore, develop a number of propositions showing how the optimal policies will change with any change in these parameters. Recall that to obtain the profit maximizing policies, we need to establish that the profit function is concave so that the first order conditions do give the unique maxima. This is established in Proposition 1. All proofs of the propositions are given in Appendix A.

Proposition 1. *Under the condition of* $4\theta - \upsilon^2\psi > 0$, *the profit function given in* Eq. (8) *is concave and has unique maxima.*

We use two figures to show the variation of profits with respect to r and m, respectively. The values of the exogenous parameters used in these numerical experimentations are given below.

Parameter	Value
p	60–150
α	1000
β	10
γ	3–5.25
δ	3–5.25
ϕ	100
ψ	1–2.8
θ	15–24
ν	0.2

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We also use these values of the exhaustive numerical analysis reported briefly in Section 5. Fig. 1 shows the variation of the profit function as the return policy r is increased for a given m. The plot confirms Proposition 1 that the profit function is concave. We also see that as m is increased for a given r, profit also increases. It is also interesting to note from Fig. 1 that a higher modularity level will increase the optimal return policy. Fig. 2 shows the variation of profit when modularity level increases for a given r showing a concave function. It is interesting to note from Fig. 2 that offering full return policy may not be the optimal choice. The profit improves as the seller offers a return policy that is closer to the optimal return policy.

In the next proposition, we obtain the closed form solutions for our decision variables r and m.

Proposition 2. *Under the condition of* Proposition 1, *the optimal policies are given as follows*:

1. *The optimal return policy is given by the equation*:

$$r^* = \frac{\gamma\theta p - 2\theta\phi + \delta\upsilon\psi p + \upsilon^2\phi\psi}{\psi(4\theta - \upsilon^2\psi)}$$

2. *The optimal modularity level is given by the equation*:

$$m^* = \frac{4\delta p + \gamma\upsilon p + 2\upsilon\phi}{4\theta - \upsilon^2\psi}$$

The proposition above gives us a way to calculate the optimal policies given the values of the sensitivity parameters. We will now analyze the effect of any changes in the sensitivity parameters to the optimum policies. The objective of this study is to generate a number of managerial guidelines that can be used for making decisions regarding return policies. First, we analyze the effect of changes in the market sensitivity parameters γ, δ, and ψ.

Proposition 3. *In the market where customer demand is more sensitive to the return policy* (*increasing* γ), *the seller will offer more generous return policy and simultaneously the optimum modularity level should be increased.*

Let us apply this insight to the computer industry, say to Dell and Compaq. Traditionally, Compaq sold through retail stores and Dell always have direct channel only. Direct channel customers are assumed to be more sensitive to return policy because of the lack of opportunity for inspection. Therefore it can be assumed that γ is high for Dell compared to Compaq. The policy obtained from Proposition 3 is that the firm should offer product with increased modularity. This is what we see with Dell. The interesting managerial insight here is that a higher γ is desirable for the firm because it increases demand and also profitability.

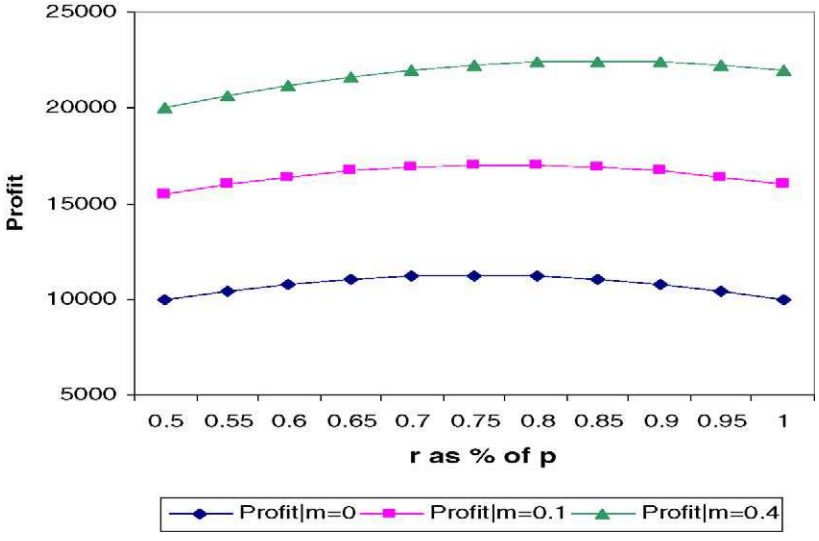


Fig. 1. Variation of profit with return policy for a given modularity level.

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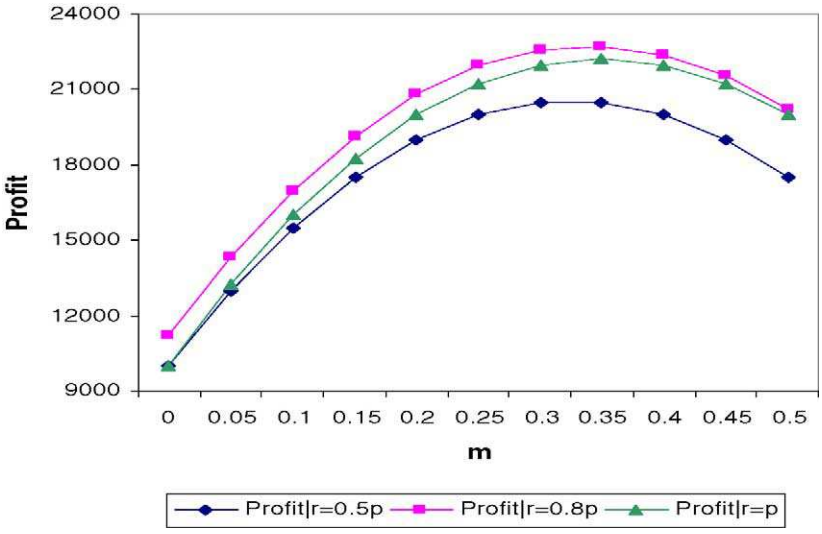


Fig. 2. Variation of profit with modularity level for a given return policy.

The increase in optimal m can be explained as follows. Since the product is customized, offering a generous return policy without increasing the modularity level will increase the cost of returning. Thus an increase in return policy should be matched by an increase in modularity level. The managerial guidelines here is that it is worthwhile for the manager to employ any possible means to influence the market to increase γ. For example, the firm can use advertising and/or other promotional means to make customers aware of its generous return policy and therefore can see an increase in γ. A similar argument also mentioned by Padmanabhan and Png (1995) that "a manufacturer that accepts return will invest in advertising and promotion."

Proposition 4. *In the market where the demand is increasingly more sensitive to the modularity level (increasing δ), the optimum modularity level will increase and the seller should simultaneously offer more generous return policy.*

In Proposition 4, we study the effect of demand sensitivity to the modularity level on our optimal policies. In this regard, let us compare two products with different modularity index. Note that a computer offered by Dell is modular while a custom made suit is not. So we can assume that computer will have a higher δ. Proposition 4 says that the optimal policy will be a higher r in conjunction with higher m. We see that Dell offers full refund while Baron Boutique (as seen in their website) allows only partial refund for their suits. Offering more generous return policy in this case could be interpreted as showing good faith in customer, as Padmanabhan and Png (1997) put it "Returns Policies: Make Money by Making Good." Once again, we see the interrelationship between the optimal policies regarding the two decision variables.

Proposition 5. *When the seller can decrease their product development and design costs (decreasing θ), the optimum modularity level will increase and at the same time the firm can offer more generous return policy.*

Proposition 6. *When the seller can salvage more from the returned product (increasing v), the optimal return policy and the optimum modularity level offered will increase.*

We see an interesting application of Proposition 5 in outsourcing. As we have seen earlier, one of the motivations for increased modularity is the firm's desire to efficiently outsource and thereby reduce costs (Salvador et al., 2002; Murray and Sako, 1999). This will lead to a lower θ and increased r. For Proposition 6, we use the example of IBM. The website of their European Division mentions that their

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"new" product contains used parts, showing that they recover a higher salvage value (higher ν) from their used products. Our proposition shows that this should be associated with higher r and what we see is that IBM return policy is more generous than others, e.g. Dell. This result is also consistent with the results from Davis et al. (1998) as they show that generous return policy is favorable when the retailer can obtain high salvage value for returned merchandise.

Proposition 7. *If there is an increase in the exogenous market price, the seller would be able to offer a more generous return policy and should design to a higher modularity level.*

Note that in our model, price is exogenously fixed and not a decision variable. A higher price should be associated with higher r and higher m. These results are consistent with previous research in the area. Padmanabhan and Png (1997) found that a higher price is associated with offering a return policy. They have also reported that McKesson, a distributor in health and beauty products, give an option to retailer to have a generous return policy while charging higher wholesale price. Pasternack (1985) where he shows that an increase in return allowances should be balanced with increase in the price. Moses and Seshadri (2000) study channel coordination and show that it can be better achieved with cost sharing than quantity discount. Their result is similar to Pasternack (1985) where the coordination is due to the producer extending a partial credit for unsold goods. We will show this result using our numerical analysis in the next section.

5. Numerical analysis

In the previous section, we have obtained a number of analytical results regarding the form of optimal return policy and optimal modularity level. In this section, we will give a brief report of the extensive numerical analysis we carried out. In this analysis, we find, using numerical means, the effect of any change in the sensitivity parameters to the decision variables. The objective of this numerical analysis is three-fold. First, it verifies the analytical results obtained in Section 4. Second, we are able to derive much deeper insights into the strategies which were not possible analytically. These will lead to useful managerial guidelines for strategic planning. Third, we recognize the fact that many of the results we obtain are predicated on the values of the parameters in the model. Numerically, we explore sensitivity analysis to see the effects of changing those parameter values on the optimal strategies. The parameters used in this set of sensitivity analysis are demand sensitivity to the return policy (γ), demand sensitivity to the modularity level (δ), product development and design costs parameter (θ), rate of return with respect to the refund amount (ψ), and salvage value parameter (ν). As price is treated as an exogenous in our model, we also carried out sensitivity analysis on price to find out its effect to the optimum policies.

We find the effect of changes of these parameters on the following resultant variables: return policy (r), modularity level (m), demand (D), returned quantity (R), and profit (π). All these results are reported in a table form below. We also have extensive graphical representations of these results which are not included here, but can be obtained from the authors.

Variables	Parameters					
	γ	δ	ψ	θ	ν	ρ
Return policy, r	+	+	−	−	+	+
Modularity level, m	+	+	+	−	+	+
Demand, D	+	+	−	−	+	−
Returned quantity, R	+	+	+	−	+	+
Profit, π	+	+	−	−	+	−

From the table above, we see that all the decision variables, namely return policy and modularity level would increase when γ, δ, and ν increased. We also see that demand and returned quantity increased. Though the increase in returned quantity actually increases costs, we see that it is more than offset by the increased revenue due to demand increase, and therefore, profit increases. So it is desirable to have a higher γ and δ. The managerial insight here is that effective advertisement can be used to increase the awareness and benefit of the generous return policy and the increased possibility of customization due to modularity. This will in turn influence γ and δ to the right direction.

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An interesting and not so obvious result was found when rate of return with respect to the refund amount (ψ) is changed. If ψ is decreased, the two decision variables move in the opposite direction; r increases and m decreases. This is a result which is different from the results obtained for the other three parameters. When customers are less inclined to return than obviously the firm can offer a more generous return policy. But at the same time, the less costly option of lower modularity can be resorted to. All three performance measures namely demand, returned quantity, and profit improved. Again, the managerial insight here is that the manager can use quality to induce customers to keep and use the product rather than return it. InternetRetailer.com cited that Linksys reported 60% decrease in their router's return rate by providing more information to consumers on what cables they needed to hook up the routers.

A very important managerial insight is obtained by experimenting on product development and design costs parameter (θ). If a manager can use operational means to reduce the cost structure, the firm can offer more generous return policy and higher modularity level and we will see all three performance measures increase. So far, as price (p) is concerned, we study what happens if there is increased competition in the market pressuring the price to go down. The firm would need to counter the loss of revenue by reducing both r and m to reduce costs. So there is a complex interplay here with demand increasing due to price decrease but is sacrificed somewhat by lowering r and m. The good news is that the net effect of using the optimal policies given by our model is an increase in demand and also in profit.

6. Conclusions and further research

Offering a return policy for a build-to-order product has always been a challenge for the firms. The customized nature of a BTO product makes it difficult to salvage any value out of the returned products and thus is not encouraging for the firm. In this paper, we propose a solution to this dilemma by suggesting the use of modularity in the product design. To the best of our knowledge, this is the first time the design modularity and the return policy are jointly optimized and the interrelationship between these two policy variables has been examined. Other novelty in our model includes using modularity index as a decision variable. Our model is also novel in proposing design modularity as a means of achieving generous and economically viable return policy for BTO products. Our model analyzes the effect of modularity and return policy (two decision variables in our model) on the product demand, amount returned, and profit. We also discuss ways to operationalize the modularity of the product design.

The trade-off in our model is that a generous return policy and or a higher modularity level will increase the revenue due to an increase in the sales but at the same time, will increase the cost of returned merchandise and cost of product design. We obtained closed form optimal solutions for our two decision variables: refund amount and modularity level. Our model should be seen as a vehicle to get insights into what directions the optimal polices should take rather than getting an absolute value for the decision variables. Through our analytical results and numerical analysis, we presented a number of insights into how a manager can influence these parameters using marketing and operational strategy variables to obtain the desired optimum directions for the firm's decision variables and get the benefit of ripple effects to increase the optimum profit.

A number of future research areas can be identified from the basic model we presented here. We use price as a given exogenous variables, and used sensitivity analysis to see how different price levels affect the optimal policies. In subsequent models, price can be used as a decision variable. Market competition has not been considered in our model. Our model can be extended to incorporate competition explicitly, as a duopoly in the simplest case. Also uncertainty can be incorporated in both the demand function and the return function. In the case of return function, one can use probabilistic return function as customers may decide not to return an item even if they could.

Appendix A

A.1. Proof for Proposition 1

Hessian matrix for our profit function as stated in Eq. (8) is $\begin{bmatrix} -4\psi & \upsilon\psi \\ \upsilon\psi & -\theta \end{bmatrix}$. The associated H_1 are -4ψ

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and $-\theta$, and H_2 is $\psi(4\theta - \upsilon^2\psi)$ where H_1 and H_2 are principal minors. The profit function is a strictly concave function if and only if all nonzero principal minors have the same sign as $(-1)^k$ where k is leading principal minor. This means that we need $H_1 < 0$ and $H_2 > 0$. Here we have $H_1 < 0$ as ψ and $\theta > 0$. So if $H_2 = \psi(4\theta - \upsilon^2\psi) > 0$ then Eq. (8) is strictly concave. □

A.2. Proof for Proposition 2

By taking the first derivative of profit with respect to return policy (r) and modularity level (m) and setting each equal to zero, we have:

$$\frac{\partial \pi}{\partial r} = p\gamma - 2\phi - 3r\psi - \psi(r - \upsilon m) = 0 \quad \text{(A.1)}$$

$$\frac{\partial \pi}{\partial m} = p\delta - m\theta + \upsilon(\phi + \psi r) = 0 \quad \text{(A.2)}$$

After some algebraic manipulations, one can obtain optimal return policy and optimal modularity level in terms of market sensitivity parameters by using Eqs. (A.1) and (A.2). The final equations for optimal policies are given below:

$$r^* = \frac{\gamma\theta p - 2\theta\phi + \delta\upsilon\psi p + \upsilon^2\phi\psi}{\psi(4\theta - \upsilon^2\psi)}$$

$$m^* = \frac{4\delta p + \gamma\upsilon p + 2\upsilon\phi}{4\theta - \upsilon^2\psi}$$

□

A.3. Proof for Proposition 3

Differentiating r^* and m^* given in Proposition 2, each with respect to the return sensitivity (γ), we obtain $\frac{\partial r^*}{\partial \gamma} = \frac{p\theta}{\psi(4\theta - \upsilon^2\psi)}$ and $\frac{\partial m^*}{\partial \gamma} = \frac{p\upsilon}{4\theta - \upsilon^2\psi}$. Since the denominators are positive and the numerator are also positive in both cases, we have $\frac{\partial r^*}{\partial \gamma} > 0$ and $\frac{\partial m^*}{\partial \gamma} > 0$. □

A.4. Proof for Proposition 4

Taking derivative of r^* and m^* given in Proposition 2, each with respect to the modularity level sensitivity (δ), we obtain $\frac{\partial r^*}{\partial \delta} = \frac{p\upsilon}{4\theta - \upsilon^2\psi}$ and $\frac{\partial m^*}{\partial \delta} = \frac{4p}{4\theta - \upsilon^2\psi}$. Since the denominators are positive and the numerators are also positive, we have both $\frac{\partial r^*}{\partial \delta} > 0$ and $\frac{\partial m^*}{\partial \delta} > 0$. □

A.5. Proof for Proposition 5

Differentiating r^* and m^* given in Proposition 2, each with respect to θ and rearranging the results, we obtain $\frac{\partial r^*}{\partial \theta} = -\frac{\upsilon(4p\delta + p\gamma\upsilon + 2\upsilon\phi)}{(4\theta - \upsilon^2\psi)^2}$ and $\frac{\partial m^*}{\partial \theta} = -\frac{4(4p\delta + p\gamma\upsilon + 2\upsilon\phi)}{(4\theta - \upsilon^2\psi)^2}$. Since the denominator are positive and the numerator are negative in both cases, we have $\frac{\partial r^*}{\partial \theta} < 0$ and $\frac{\partial m^*}{\partial \theta} < 0$. □

A.6. Proof for Proposition 6

Differentiating r^* and m^* given in Proposition 2, each with respect to υ and rearranging the results, we obtain $\frac{\partial r^*}{\partial \upsilon} = \frac{4\theta\upsilon\phi + p(4\delta\theta + 2\gamma\theta\upsilon + \delta\upsilon^2\psi)}{(4\theta - \upsilon^2\psi)^2}$ and $\frac{\partial m^*}{\partial \upsilon} = \frac{4\theta(p\gamma + 2\phi) + \upsilon\psi(8p\delta + p\gamma\upsilon + 2\upsilon\phi)}{(4\theta - \upsilon^2\psi)^2}$. Since both of the denominator and the numerator are positive, we can conclude that $\frac{\partial r^*}{\partial \upsilon} > 0$ and $\frac{\partial m^*}{\partial \upsilon} > 0$. □

A.7. Proof for Proposition 7

Taking the derivative of r^* and m^* given in Proposition 2, each with respect to υ, we obtain $\frac{\partial r^*}{\partial p} = \frac{\gamma\theta + \delta\upsilon\psi}{\psi(4\theta - \upsilon^2\psi)}$ and $\frac{\partial m^*}{\partial p} = \frac{4\delta + \gamma\upsilon}{4\theta - \upsilon^2\psi}$. Since both of the denominator and the numerator are positive, we can conclude that $\frac{\partial r^*}{\partial \upsilon} > 0$ and $\frac{\partial m^*}{\partial \upsilon} > 0$. □

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